

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – X

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Class: TYIT	Batch: 2
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Data Preparation and Transformation using Tableau Prep

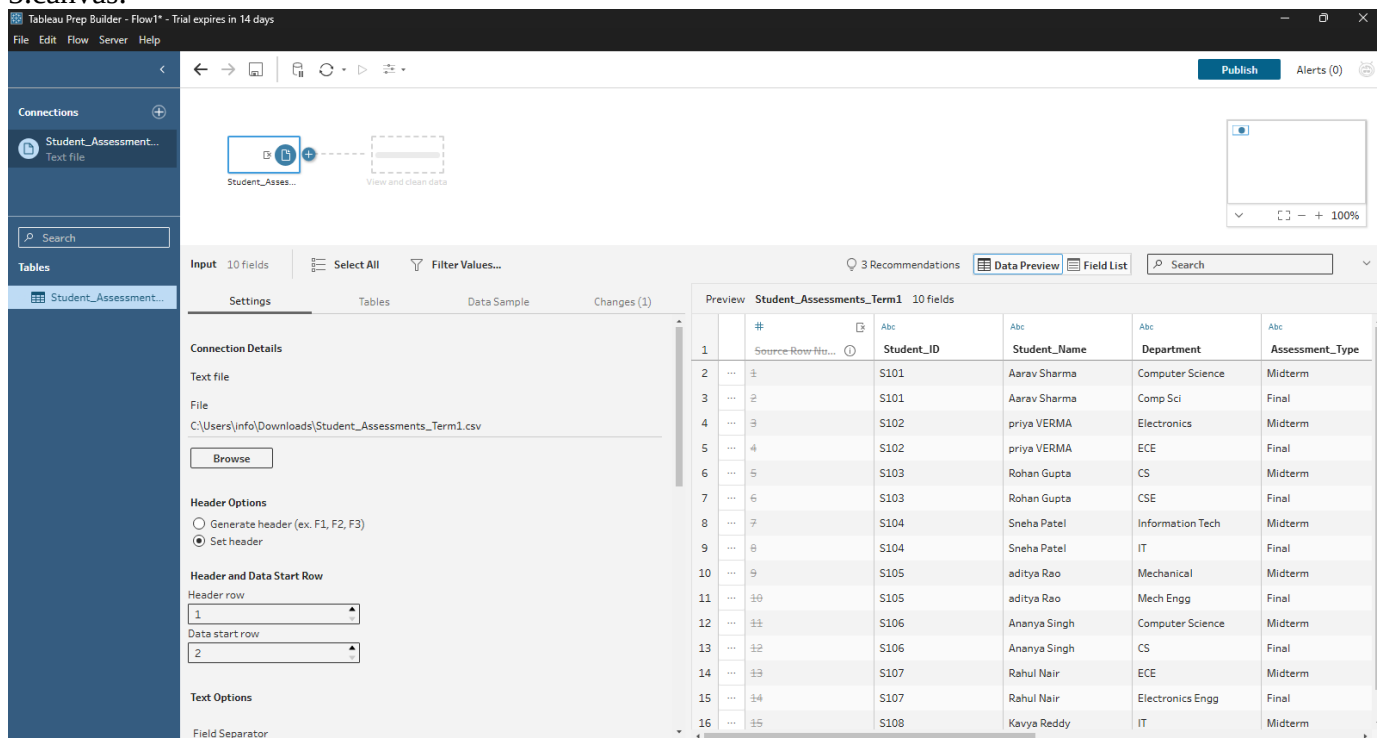
A) Connect and load data into Tableau Prep

Step 1: Open Tableau Prep Builder

1. Launch Tableau Prep Builder on your machine.
2. On the start screen, locate the Connections pane on the left-hand side.

Step 2: Load the First Dataset (Student_Assessments_Term1.csv)

1. In the Connections pane, click the + (Add connection) button.
2. In the list of file formats, select Text file.
3. Browse to the folder where you saved the CSVs, select Student_Assessments_Term1.csv, and click Open.
4. An Input Step icon named Student_Assessments_Term1 will appear on the Flow canvas.
5. canvas.



5. In the bottom Input configuration pane, verify:

- o Field Separator: Comma (,)
- o First row contains header: Checked

Step 3: Load the Second and Third Datasets

1. In the top-left Connections pane, click the + (Add connection) button again.
2. Select Text file, browse and select Student_Assessments_Term2.csv, and click Open.
3. Click + (Add connection) once more -> select Text file -> choose Course_Credit_Master.csv -> click Open.
4. You will now see three separate Input icons on your Flow canvas.

Step 4: Add an Initial Clean Step to Term 1

1. On the Flow canvas, hover your mouse over the Student_Assessments_Term1 input icon.
2. Click the small + icon that appears to its right.
3. Select Clean Step (the step with the cleaning tool icon).

The screenshot shows the Alteryx interface. On the left, the 'Connections' pane lists 'Student_Assessment...' (Text file), 'Student_Assessment...' (Text file), and 'Course_Credit_Master...' (Text file). Below it, the 'Tables' pane shows 'Course_Credit_Master'. The main canvas displays a flow with 'Student_Assessments_Term1' and 'Course_Credit_Master' inputs. A context menu is open over the 'Student_Assessments_Term1' input, showing the 'Add' menu with 'Clean Step' selected. The 'Clean Step' menu includes options: New Rows, Aggregate, Pivot, Join, Union, Script, Prediction, Output, and Insert Flow. The bottom pane shows the 'Data Preview' for 'Course_Credit_Master' with 5 fields: #, Subject_Code, Subject_Title, Credit_Hours, and Passing_Marks. The data is as follows:

#	Subject_Code	Subject_Title	Credit_Hours	Passing_Marks
1	CS101	Data Structures	4	40
2	CS201	Algorithms	4	40
3	EC101	Digital Logic	3	40
4	EC201	Signals & Systems	4	40
5	IT101	Database Systems	4	40
6	IT201	Cloud Computing	4	40

4. The Profile Pane will open at the bottom, displaying interactive profile cards for each of the 9 columns in Term 1.

The screenshot shows the Alteryx interface with the 'Clean 1' step connected to 'Student_Assessments_Term1'. The 'Profile Pane' is open at the bottom, displaying interactive profile cards for each of the 9 columns in Term 1. The cards show data distribution and statistics for columns like Student_ID, Student_Name, Department, Assessment..., Exam_Date, Subject_Code, and Marks_Obtain....

Column	Student_ID	Student_Name	Department	Assessment...	Exam_Date	Subject_Code	Marks_Obtain...
1	S101	Aarav Sharma	Comp Sci	Final	15-09-2025	CS101	35
2	S102	aditya Rao	Computer Science	Midterm	20-11-2025	EC101	55
3	S103	Ananya Singh	CS		2025.09.15	IT101	75
4	S104	Kavya Reddy	CSE		2025.11.20	ME101	95
5	S105	priya VERMA	ECE		2025/09/15		
6	S106	Rahul Nair	Electronics		2025/11/20		
7	S107	Rohan Gupta	Electronics Engg				
8	S108	Sneha Patel	Info Tech				
9			Information Tech				

B) Clean and profile data

Step 1: Inspect the Profile Pane

1. Click on the Clean 1 step connected to Student_Assessments_Term1.
2. Look at the cards displayed in the Profile Pane:
 - o Each card represents one column.
 - o Notice that Student_Name has multiple representations of the same student due to extra spaces and mixed casing (e.g., 'Aarav Sharma ' vs 'Aarav Sharma', 'priya VERMA', 'aditya Rao ').
 - o Notice that Exam_Date is recognized as a string (Abc) due to mixed separators (/ , - , .).

Step 2: Clean Whitespace and Standardize Letter Casing

- On the Student_Name profile card:
 - Click the More options (...) icon at the top right of the card.
 - Go to Clean --> select Trim Spaces (this strips all leading and trailing whitespace).
 - Click More options (...) again --> Clean --> select Make Uppercase (or Make Lowercase / standard proper formatting).

The screenshot shows the 'Clean 1' card in the data tool. The context menu is open, and the 'Clean' submenu is visible. The 'Trim Spaces' option is highlighted. The data table below shows the following data:

Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained	Max
Aarav Sharma	Comp Sci	Final	15-09-2025	CS101	35	100
Aditya Rao	Computer Science	Midterm	20-11-2025	EC101	55	
Ananya Singh	CS		2025-09-15	IT101	75	
Kavya Reddy	CSE		2025-11-20	ME101	95	
Priya Verma	ECE		2025/09/15			
Rahul Nair	Electronics		2025/11/20			
Rohan Gupta	Electronics Engg					
Sneha Patel	Info Tech					
	Information Tech					
	IT					
	Mech Engg					
	Mechanical					

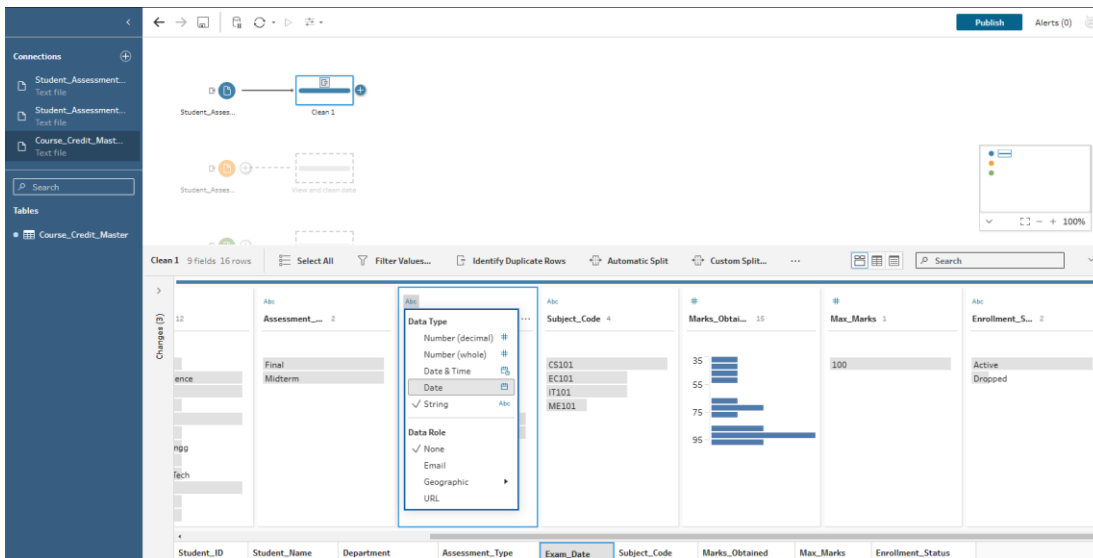
2. The card updates immediately, consolidating duplicate whitespace variations.

The screenshot shows the 'Clean 1' card updated. The data table now shows the following data:

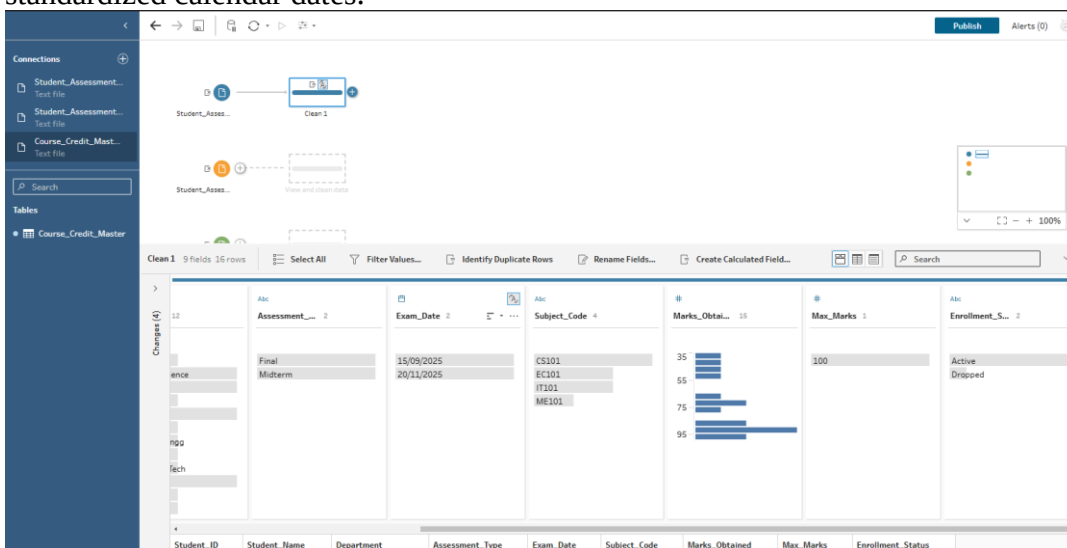
Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained	Max
AARAV SHARMA	Comp Sci	Final	15-09-2025	CS101	35	100
ADITYA RAO	Computer Science	Midterm	20-11-2025	EC101	55	
ANANYA SINGH	CS		2025-09-15	IT101	75	
KAVYA REDDY	CSE		2025-11-20	ME101	95	
PRIYA VERMA	ECE		2025/09/15			
RAHUL NAIR	Electronics		2025/11/20			
ROHAN GUPTA	Electronics Engg					
SNEHA PATEL	Info Tech					
	Information Tech					
	IT					
	Mech Engg					
	Mechanical					

Step 3: Convert Exam_Date to Date Data Type

- Locate the Exam_Date profile card.
- Click the data type icon (currently showing Abc) directly above the column name.
- Select Date from the dropdown menu.

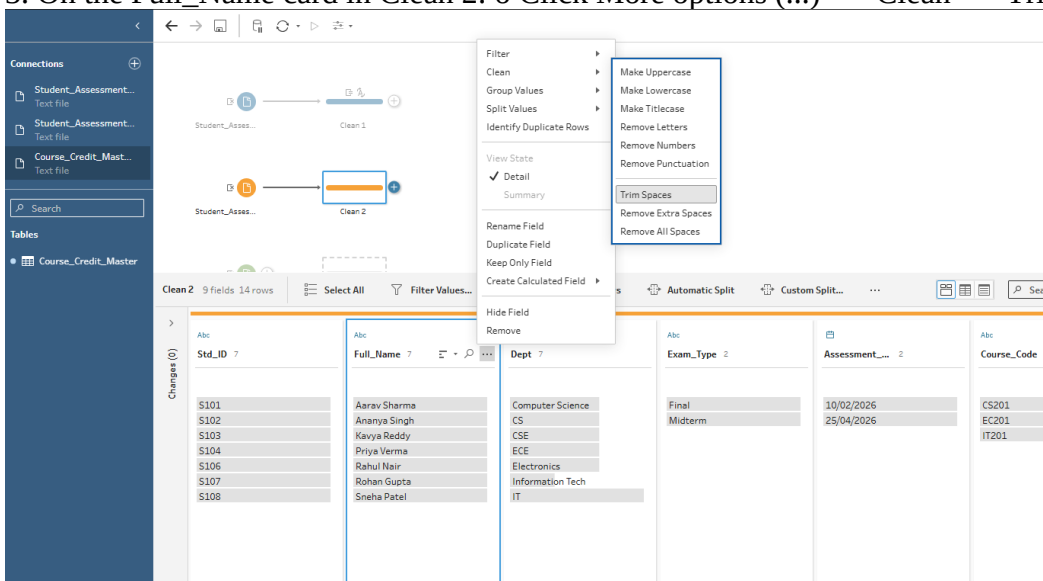


4. Tableau Prep automatically parses the various date formats (2025/09/15, 15-09-2025, 2025.11.20) into standardized calendar dates.



Step 4: Clean the Second Dataset (Student_Assessments_Term2.csv)

1. On the Flow canvas, hover over the Student_Assessments_Term2 input icon.
2. Click + --> select Clean Step (creates Clean 2).
3. On the Full_Name card in Clean 2: o Click More options (...) --> Clean --> Trim Spaces.



4. On the Assessment_Date card: Click the Abc icon --> change the data type to **Date**.

The screenshot shows the Power BI interface with the 'Clean 2' step selected. The 'Assessment_Date' field is highlighted, and the 'Data Type' dropdown menu is open, showing options like Number (decimal), Number (whole), Date & Time, Date, String, Data Role, None, Email, Geographic, and URL. The 'Date' option is selected.

Std_ID	Full_Name	Dept	Exam_Type	Assessment_Date	Course_Code	Score	Total_Possible	Enrollment_Status
S101	Aarav Sharma	Computer Science	Final		CS201	50		
S102	Ananya Singh	CS	Midterm		EC201	60		
S103	Kavya Reddy	CSE			IT201	70		
S104	Priya Verma	ECE				75		
S106	Rahul Nair	Electronics				82		
S107	Rohan Gupta	Information Tech				88		
S108	Sneha Patel	IT				90		
						94		
						95		
						96		
						98		
						100		

The screenshot shows the Power BI interface with the 'Clean 2' step selected. The 'Assessment_Date' field is highlighted, and the 'Data Type' dropdown menu is open, showing options like Number (decimal), Number (whole), Date & Time, Date, String, Data Role, None, Email, Geographic, and URL. The 'Date' option is selected.

Std_ID	Full_Name	Dept	Exam_Type	Assessment_Date	Course_Code	Score	Total_Possible	Enrollment_Status
S101	Aarav Sharma	Computer Science	Final	10/02/2026	CS201	50		
S102	Ananya Singh	CS	Midterm	25/04/2026	EC201	60		
S103	Kavya Reddy	CSE			IT201	70		
S104	Priya Verma	ECE				75		
S106	Rahul Nair	Electronics				82		
S107	Rohan Gupta	Information Tech				88		
S108	Sneha Patel	IT				90		
						94		
						95		
						96		
						98		
						100		

C| Group and standardize values

Step 1: Inspect the Department Field in Term 1 (Clean 1)

1. Click on the Clean 1 step connected to Student_Assessments_Term1.
2. Locate the Department card in the Profile Pane.
3. Observe that there are 12 distinct entries representing only 4 actual academic departments:
 - o Computer Science (Computer Science, Comp Sci, CS, CSE)
 - o Electronics (Electronics, ECE, Electronics Engg)
 - o Information Technology (Information Tech, IT, Info Tech)
 - o Mechanical Engineering (Mechanical, Mech Engg)

Connections: Student_Assessment... Text file, Student_Assessment... Text file, Course_Credit_Mast... Text file

Tables: Course_Credit_Master

Clean 1 9 fields 16 rows

Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained
S101	AARAV SHARMA	Comp Sci	Final	15/09/2025	CS101	35
S102	ADITYA RAO	Computer Science	Midterm	20/11/2025	EC101	55
S103	ANANYA SINGH	CS			IT101	75
S104	KAVYA REDDY	CSE			ME101	95
S105	PRIYA VERMA	ECE				
S106	RAHUL NAIR	Electronics				
S107	ROHAN GUPTA	Electronics Engg				
S108	SNEHA PATEL	Info Tech				
		Information Tech				
		IT				
		Mech Engg				
		Mechanical				

Step 2: Group Values Using Pronunciation / Spelling or Manual Selection

1. On the Department card, click the More options (...) menu at the top right.
2. Go to Group Values -->select Manual Selection (or Pronunciation / Common Characters).

Connections: Student_Assessment... Text file, Student_Assessment... Text file, Course_Credit_Mast... Text file

Tables: Course_Credit_Master

Clean 1 9 fields 16 rows

Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained
S101	AARAV SHARMA	Comp Sci	Final	15/09/2025	CS101	35
S102	ADITYA RAO	Computer Science	Midterm	20/11/2025	EC101	55
S103	ANANYA SINGH	CS			IT101	75
S104	KAVYA REDDY	CSE			ME101	95
S105	PRIYA VERMA	ECE				
S106	RAHUL NAIR	Electronics				
S107	ROHAN GUPTA	Electronics Engg				
S108	SNEHA PATEL	Info Tech				
		Information Tech				
		IT				
		Mech Engg				
		Mechanical				

3. In the Grouping editor:

- o Click on Computer Science in the left column. In the right-hand panel, check the boxes for Comp Sci, CS, and CSE.

Connections: Student_Assessment... Text file, Student_Assessment... Text file, Course_Credit_Mast... Text file

Tables: Course_Credit_Master

Clean 1 9 fields 16 rows

Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained
S101	AARAV SHARMA	Comp Sci	Final	15/09/2025	CS101	35
S102	ADITYA RAO	Computer Science	Midterm	20/11/2025	EC101	55
S103	ANANYA SINGH	CS			IT101	75
S104	KAVYA REDDY	CSE			ME101	95
S105	PRIYA VERMA	ECE				
S106	RAHUL NAIR	Electronics				
S107	ROHAN GUPTA	Electronics Engg				
S108	SNEHA PATEL	Info Tech				
		Information Tech				
		IT				
		Mech Engg				
		Mechanical				

o Click on Electronics (or rename the master value to Electronics & Communication). Check the boxes for ECE and Electronics Engg.

The screenshot shows the 'Group Values by Manual Selection' dialog box. The 'Department' list on the left includes 'Computer Science', 'Electronics', 'Electronics & Communication' (selected), 'Info Tech', 'Information Tech', 'IT', 'Mech Engg', and 'Mechanical'. The 'Assessment_Type' list on the right includes 'Final' and 'Midterm' (both selected). The 'Student_Assessment...' table in the background shows columns for 'Student_ID', 'Student_Name', 'Department', 'Assessment_Type', 'Exam_Date', 'Subject_Code', 'Marks_Obtained', 'Max_Marks', and 'Enrollment_Status'.

o Click on Information Tech (or rename to Information Technology). Check the boxes for IT and Info Tech.

The screenshot shows the 'Group Values by Manual Selection' dialog box. The 'Department' list on the left includes 'Computer Science', 'Electronics & Communication', 'Information Technology' (selected), 'Mech Engg', and 'Mechanical'. The 'Assessment_Type' list on the right includes 'Final' and 'Midterm' (both selected). The 'Student_Assessment...' table in the background shows columns for 'Student_ID', 'Student_Name', 'Department', 'Assessment_Type', 'Exam_Date', 'Subject_Code', 'Marks_Obtained', 'Max_Marks', and 'Enrollment_Status'.

o Click on Mechanical (or rename to Mechanical Engineering). Check the box for Mech Engg.

The screenshot shows the 'Group Values by Manual Selection' dialog box. The 'Department' list on the left includes 'Computer Science', 'Electronics & Communication', 'Information Technology', and 'Mechanical Engineering' (selected). The 'Assessment_Type' list on the right includes 'Final' and 'Midterm' (both selected). The 'Student_Assessment...' table in the background shows columns for 'Student_ID', 'Student_Name', 'Department', 'Assessment_Type', 'Exam_Date', 'Subject_Code', 'Marks_Obtained', 'Max_Marks', and 'Enrollment_Status'.

4. Click Done at the top right of the editor.

5. Notice that the paperclip icon appears next to grouped values, consolidating the 12 messy entries into exactly 4 clean departments.

The screenshot shows the Alteryx interface with the 'Clean 1' step selected. The 'Department' field is being cleaned, with a list of 12 messy entries on the left and a list of 4 clean departments on the right. The 'Student_Assessment...' table is visible in the background.

Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code
S101	Aarav Sharma	Comp Sci	Final	15-09-2025	CS101
S102	aditya Rao	Computer science	Midterm	20-11-2025	EC101
S103	Ananya Singh	CS		2025.09.15	IT101
S104	Kavya Reddy	CSE		2025.11.20	ME101
S105	priya VERMA	ECE		2025/09/15	
S106	Rahul Nair	Electronics and Communicat		2025/11/20	
S107	Rohan Gupta	Electronics Engg			
S108	Sneha Patel	Info Tech			
		Information Technology			
		IT			
		Mech Engg			
		Mechanical Engineering			

Step 3: Repeat Grouping on Term 2 (Clean 2)

1. Click on the Clean 2 step connected to Student_Assessments_Term2.

2. Locate the Dept card.

3. Click More options (...) -->Group Values -->Manual Selection:

o Group CS, CSE, and Computer Science into Computer Science.

o Group ECE and Electronics into Electronics.

o Group IT and Information Tech into Information Tech.

4. Click Done.

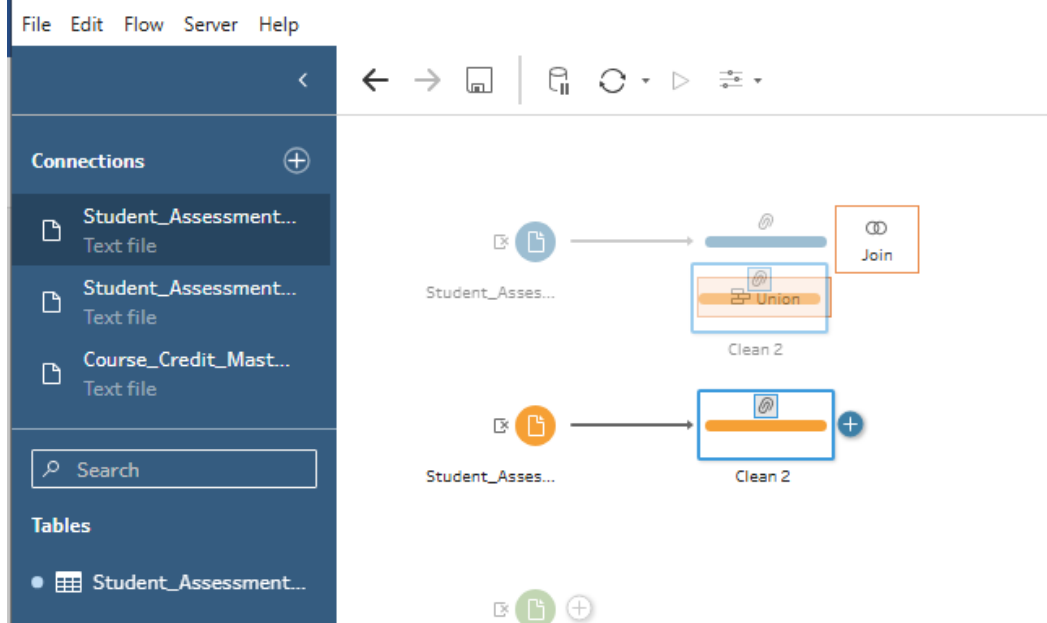
The screenshot shows the Alteryx interface with the 'Clean 2' step selected. The 'Dept' field is being cleaned, with a list of 3 messy entries on the left and a list of 3 clean departments on the right. The 'Student_Assessment...' table is visible in the background.

Std_ID	Full_Name	Dept	Exam_Type
S101	Aarav Sharma	Computer Science	Final
S102	Ananya Singh	Electronics	Midterm
S103	Kavya Reddy	Information Tech	
S104	Priya Verma		
S106	Rahul Nair		
S107	Rohan Gupta		
S108	Sneha Patel		

D| Merge mismatched fields

Step 1: Create the Union Step on the Canvas

1. On the Flow canvas, click and drag the Clean 2 step icon over to the Clean 1 step icon.



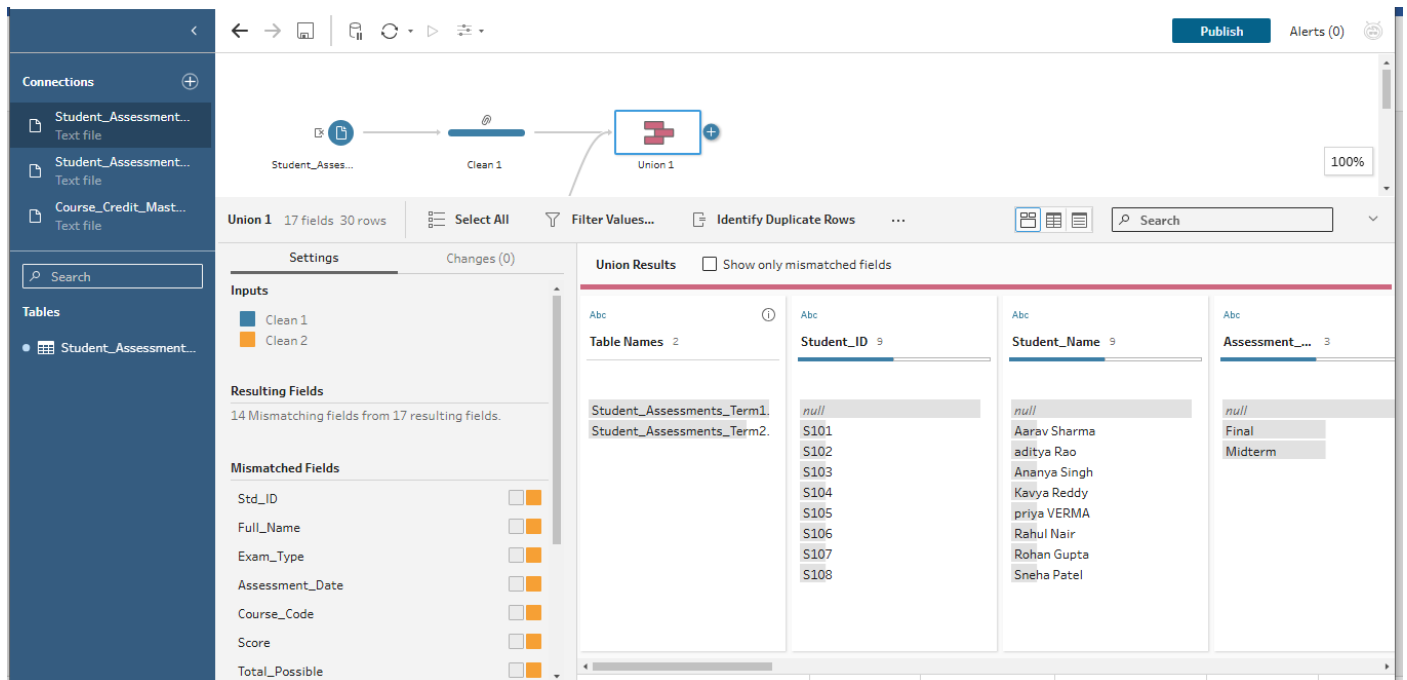
2. When the drop zone appears, hover over Union and release the mouse.
3. A new Union 1 step is created, connecting both transformation branches.

The screenshot shows the Tableau Prep interface with the 'Union 1' step created. The main canvas shows 'Clean 1' and 'Clean 2' steps connected to a 'Union 1' step. The bottom pane shows 'Union Results' with a table of data. The 'Mismatched Fields' section lists 16 mismatching fields from 18 resulting fields.

Table Names	Student_ID	Student_Name	Department
Student_Assessments_Term1	std101	Aarav Sharma	Comp Sci
Student_Assessments_Term2	std102	aditya Rao	Computer science
	std103	Ananya Singh	CS

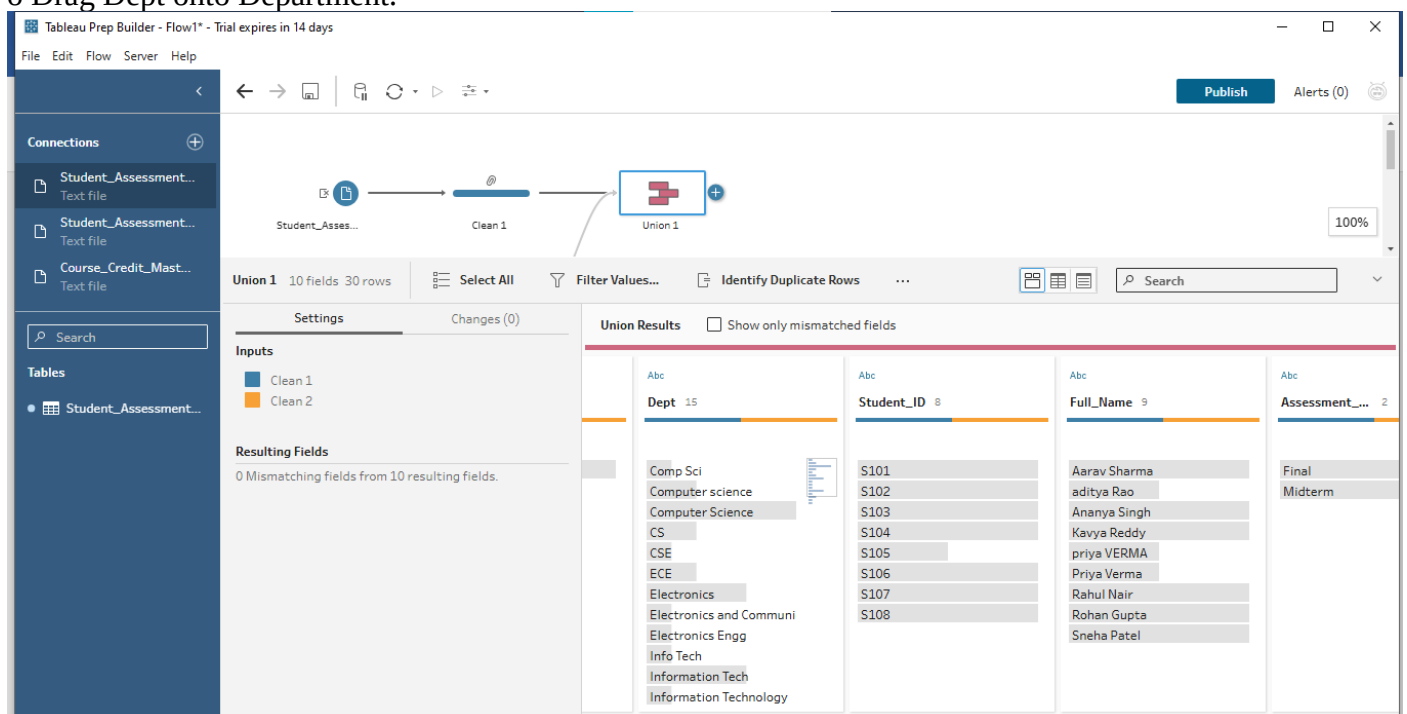
Step 2: Inspect Mismatched Fields in the Union Pane

1. Click on the Union 1 step.
2. In the bottom Union summary pane, look at the Mismatched Fields section.
3. Notice that because the two files used different naming conventions, Tableau Prep lists non-matching pairs (indicated by yellow and blue bars), such as:
 - o Student_ID (Term 1) vs Std_ID (Term 2)
 - o Student_Name (Term 1) vs Full_Name (Term 2)
 - o Department (Term 1) vs Dept (Term 2)
 - o Assessment_Type (Term 1) vs Exam_Type (Term 2)
 - o Exam_Date (Term 1) vs Assessment_Date (Term 2)
 - o Subject_Code (Term 1) vs Course_Code (Term 2)
 - o Marks_Obtained (Term 1) vs Score (Term 2)
 - o Max_Marks (Term 1) vs Total_Possible (Term 2)



Step 3: Merge the Mismatched Fields

1. In the Mismatched Fields list (or the union profile pane):
 - o Click and drag Std_ID directly on top of Student_ID to merge them into one column.
 - o Drag Full_Name onto Student_Name.
 - o Drag Dept onto Department.



Note: If You cant drag it Observe in the left pane click the dept column the hover near department column's plus icon and add the dept column into it

- o Drag Exam_Type onto Assessment_Type.
- o Drag Assessment_Date onto Exam_Date.
- o Drag Course_Code onto Subject_Code.
- o Drag Score onto Marks_Obtained.
- o Drag Total_Possible onto Max_Marks.

2. Notice that as you merge them, the mismatched count drops to 0, and each unified column displays both source colors.

File Edit Flow Server Help

Connections

- Student_Assessment... Text file
- Student_Assessment... Text file
- Course_Credit_Mast... Text file

Search

Tables

- Student_Assessment...

Union 1 10 fields 30 rows

Select All Filter Values... Identify Duplicate Rows

Settings Changes (0)

Inputs

- Clean 1
- Clean 2

Resulting Fields

0 Mismatching fields from 10 resulting fields.

Union Results

Abc

Dept 15

Comp Sci
Computer science
Computer Science
CS
CSE
ECE
Electronics
Electronics and Communi
Electronics Engg
Info Tech
Information Tech
Information Technology

Step 4: Add a Clean Step Post-Union

1. Click the + icon next to the Union 1 node.

Tableau Prep Builder - Flow1* - Trial expires in 14 days

File Edit Flow Server Help

Connections

- Student_Assessment... Text file
- Student_Assessment... Text file
- Course_Credit_Mast... Text file

Search

Tables

- Student_Assessment...

Union 1 10 fields 30 rows

Select All Filter Values... Identify Duplicate Rows

Settings Changes (0)

Inputs

Union 1

Add:

- + Clean Step
- New Rows
- Aggregate
- Pivot
- Join
- Union
- Script
- Prediction
- Output
- Insert Flow

2. Select Clean Step (creates Clean 3). In Screenshot it is showing Clean 4 because I have created clean step for Course_Credit_Master.csv too.
3. Verify that the combined dataset now contains 30 total rows across 10 unified columns (including Table Names).

File Edit Flow Server Help

Connections

- Student_Assessment... Text file
- Student_Assessment... Text file
- Course_Credit_Mast... Text file

Search

Tables

- Student_Assessment...

100%

Clean 4 10 fields 30 rows

Select All Filter Values... Identify Duplicate Rows

Search

Table Names	Enrollment_S...	Dept	Student_ID	Full_Name	Assessment
Student_Assessments_Term1	Active	Comp Sci	S101	Aarav Sharma	Final
Student_Assessments_Term2	Dropped	Computer science	S102	aditya Rao	Midterm
		Computer Science	S103	Ananya Singh	
		CS	S104	Kavya Reddy	
		CSE	S105	priya VERMA	
		ECE	S106	Priya Verma	
		Electronics	S107	Rahul Nair	
		Electronics and Communi	S108	Rohan Gupta	
		Electronics Engg		Sneha Patel	

E| Perform aggregations in Prep

Step 1: Add a Term Identifier Field in Clean 3

1. Click on the Clean 3 step (the clean step created after Union 1).
2. In the toolbar above the Profile cards, click Create Calculated Field You have to click three dots on the toolbar.
3. Set the Field Name to: Term
4. Enter the formula:
Code snippet
IF CONTAINS([Table Names], "Term1") THEN "Term 1"
ELSEIF CONTAINS([Table Names], "Term2") THEN "Term 2"
ELSE "Other"
END
4. Click Save.

Add Field

Field Name

Term

IF CONTAINS([Table Names], "Term1") THEN "Term 1"
ELSEIF CONTAINS([Table Names], "Term2") THEN "Term 2"
ELSE "Other"
END

All

Search

ABS

ACOS

AND

AREA

ASC

ASCII

ASIN

ATAN

ATAN2

AVG

BUFFER

CASE

CEILING

CHAR

CONTAINS

COS

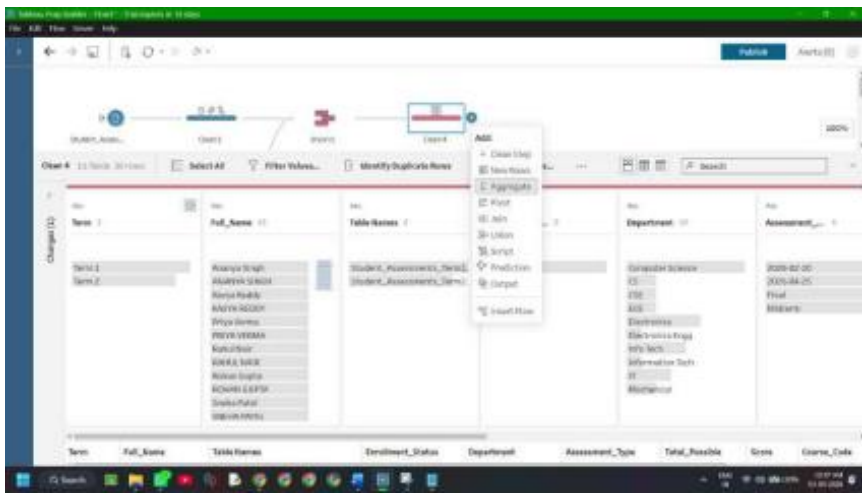
ABS(number)

Returns the absol
given number.

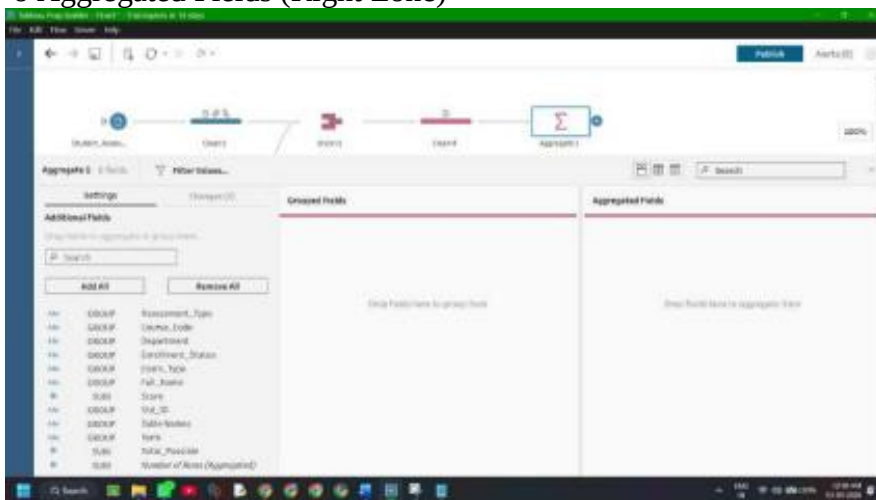
Example: ABS(-7)

Step 2: Add an Aggregate Step to the Flow Canvas

1. Hover over the Clean 3 node on the Flow canvas.
2. Click the + icon next to it.
3. Select Aggregate (represented by the summation icon Σ).



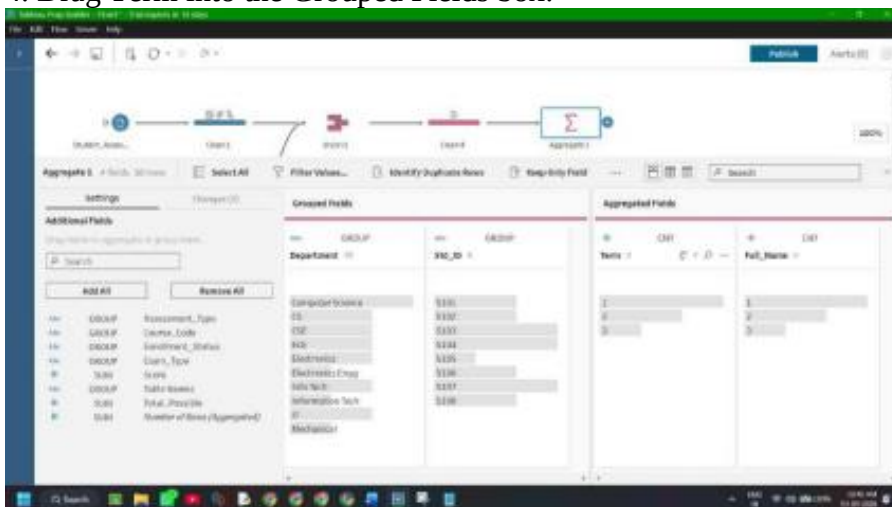
4. An Aggregate 1 step will appear, showing two drop zones in the configuration pane:
- o Grouped Fields (Left Zone)
 - o Aggregated Fields (Right Zone)



Step 3: Configure Grouped Fields (Dimensions)

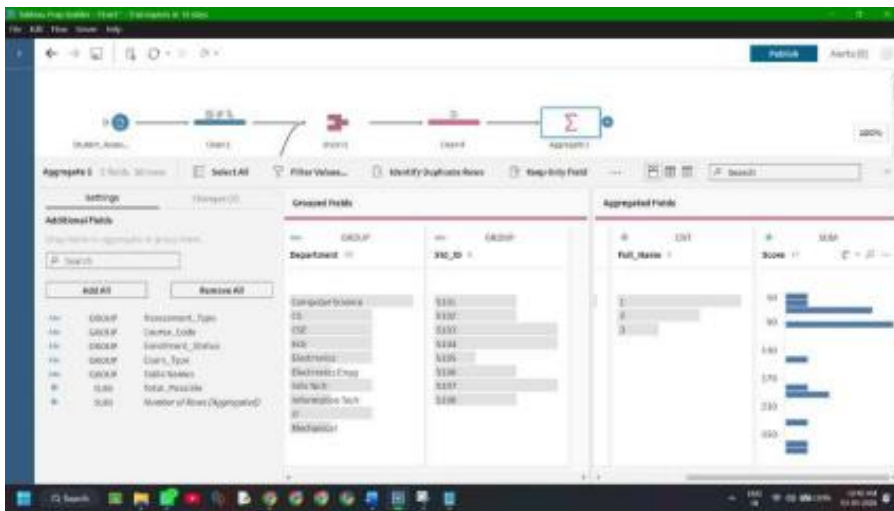
From the left-hand column list in the Aggregate step pane Your Column names could be different based on how you did previous steps of Union:

1. Drag Student_ID into the Grouped Fields box.
2. Drag Student_Name into the Grouped Fields box.
3. Drag Department into the Grouped Fields box.
4. Drag Term into the Grouped Fields box.

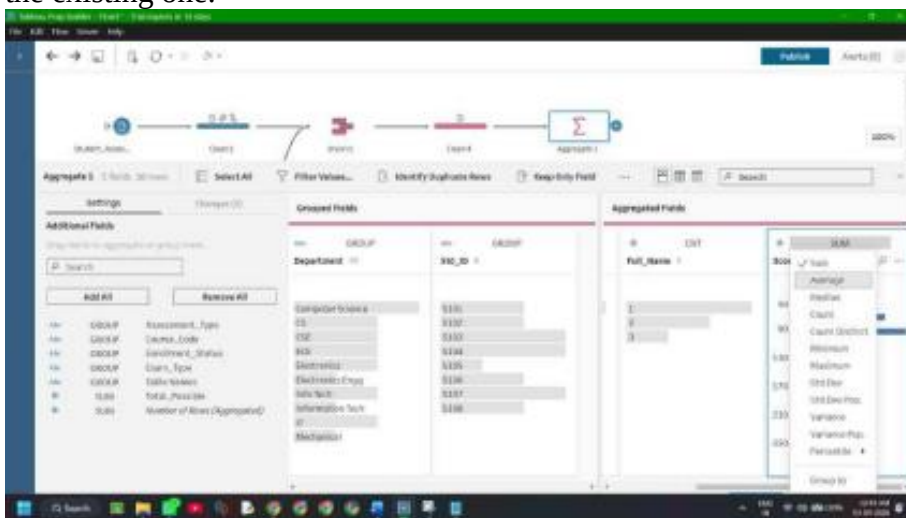


Step 4: Configure Aggregated Fields (Measures & Metrics)

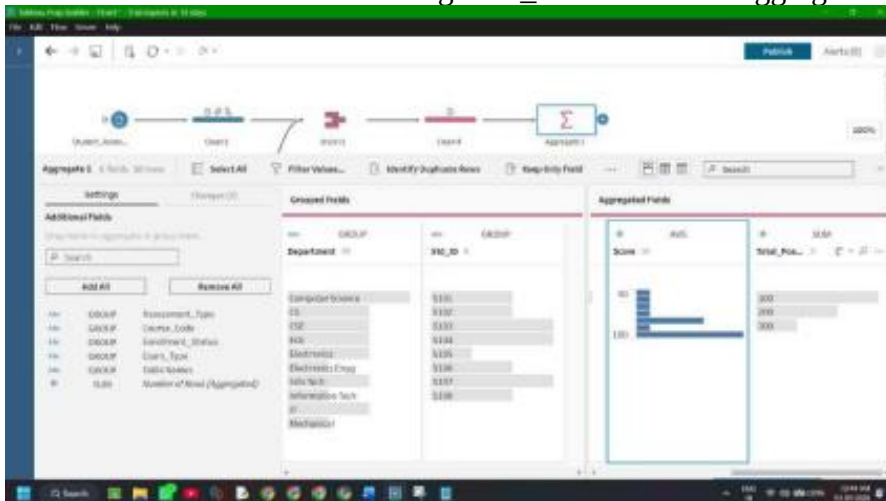
1. Total Marks: Drag Marks_Obtained into the Aggregated Fields box. (Ensure aggregation operation is set to Sum).



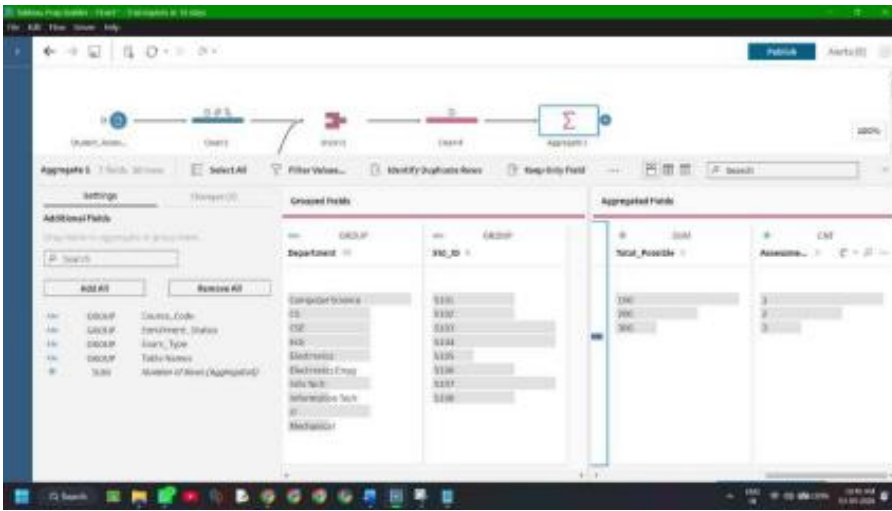
2. Average Score: Drag Marks_Obtained into the Aggregated Fields box a second time --> click on Sum --> change it to Average. There will be no field for dragging by any chance just change SUM to AVERAGE of the existing one.



2. Total Possible Marks: Drag Max_Marks into the Aggregated Fields box --> keep as Sum.



4. Assessment Count: Drag Assessment_Type (or Subject_Code) into the Aggregated Fields box --> set aggregation to Count (or Count Distinct).



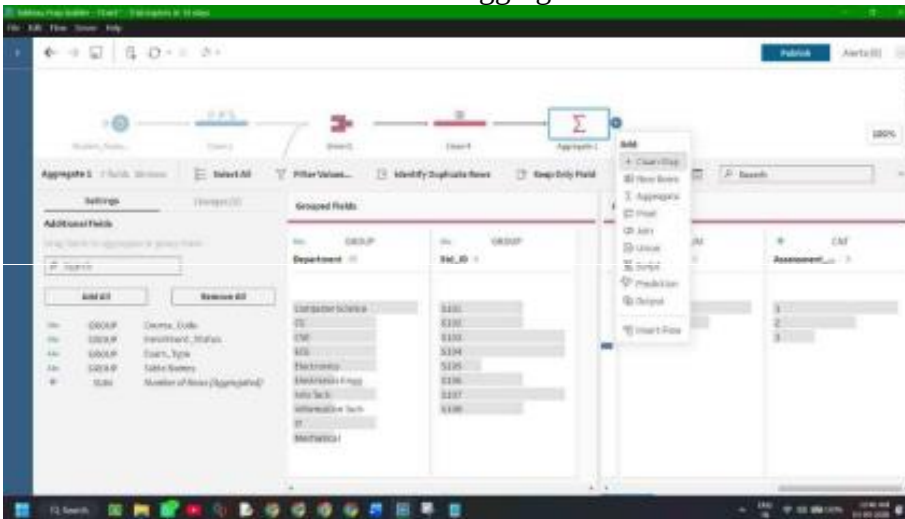
Step 5: Rename Aggregated Output Fields

In the Aggregated Fields panel, click on each field name to rename them cleanly:

- ☐ Marks_Obtained (Sum) --> Total_Marks_Obtained
- ☐ Marks_Obtained (Average) --> Average_Score
- ☐ Max_Marks (Sum) --> Total_Max_Marks
- ☐ Assessment_Type (Count) --> Assessments_Taken

Step 6: Add a Clean Step to Inspect Aggregated Output

1. Click the + icon next to the Aggregate 1 node --> select Clean Step (creates Clean4).



2. Look at the Profile Pane:

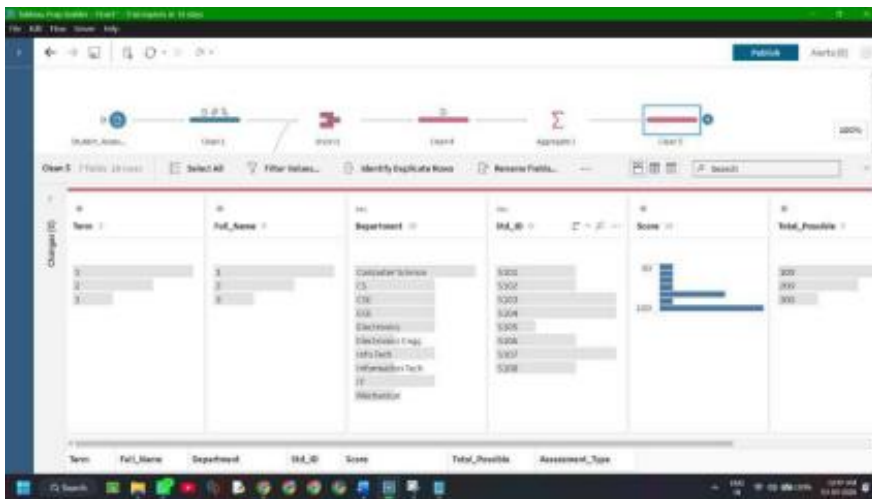
o The row count collapses from 30 individual assessment rows into 15 summary records or more (one row per student per term).

F |Apply Filters and Transformations

Step 1: Open the Transformation Step (Clean 4)

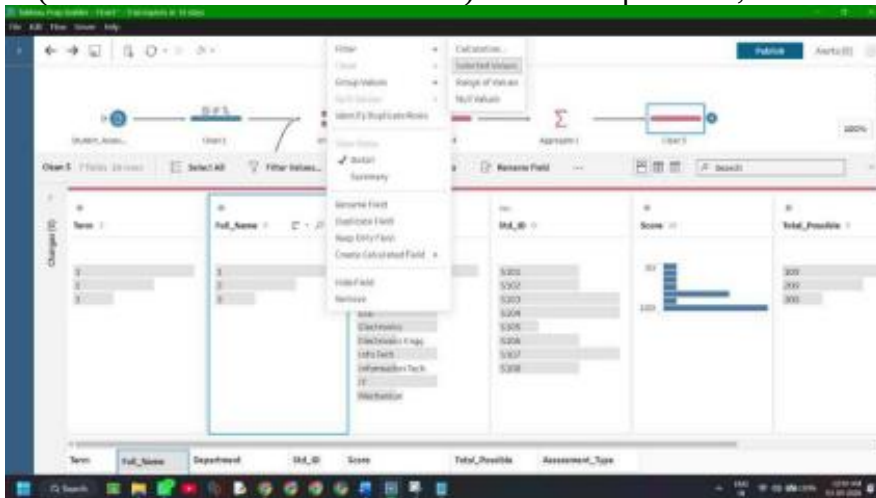
1. Click on the Clean 4 step connected after your Aggregate 1 step on the Flow canvas.

2. The Profile Pane below will show your aggregated columns: Student_ID, Student_Name, Department, Term, Total_Marks_Obtained, Average_Score, Total_Max_Marks, and Assessments_Taken. (Naming Will be different and completely depends on how you performed the Union Step)

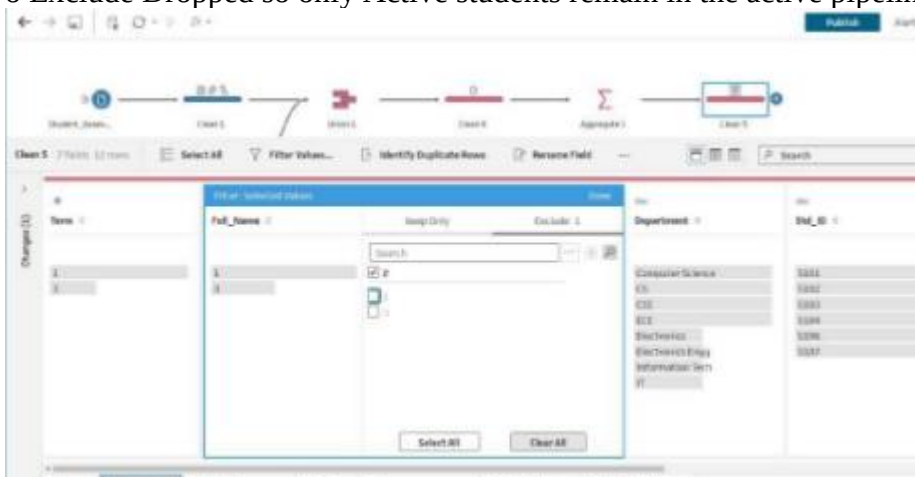


Step 2: Filter Out Inactive / Dropped Student Records

1. If Enrollment_Status is present on a profile card: o Click on the card --> click the value Dropped --> right-click and select Exclude (or click the filter icon at the top of the card and check only Active).
2. (Alternative via Filter calculation): o In the top toolbar, click Filter Values --> Selected Values....



- o Exclude Dropped so only Active students remain in the active pipeline.



Step 3: Create the Normalized Term Percentage Metric

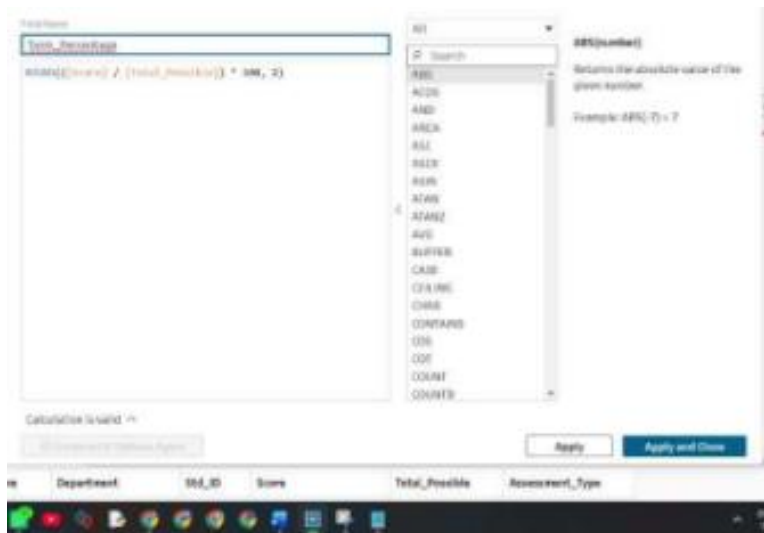
1. In the toolbar above the Profile cards, click Create Calculated Field.
2. Field Name: Term_Percentage
3. Tableau Prep Formula:

Code snippet

$\text{ROUND}([\text{Total_Marks_Obtained}] / [\text{Total_Max_Marks}] * 100, 2)$

$\text{ROUND}([\text{Score}] / [\text{Total_Possible}] * 100, 2)$ {This formula can be used if you couldn't able to rename your aggregate fields.}

4. Click Save.



Step 4: Create the Categorical Letter Grade Transformation

1. In the toolbar, click Create Calculated Field.

2. Field Name: Letter_Grade

3. Tableau Prep Formula:

Code snippet

```
IF [Term_Percentage] >= 85 THEN "A (Distinction)"
ELSEIF [Term_Percentage] >= 75 THEN "B (First Class)"
ELSEIF [Term_Percentage] >= 60 THEN "C (Second Class)"
ELSEIF [Term_Percentage] >= 40 THEN "D (Pass Class)"
ELSE "F (Fail / Re-appear)"
END
```

3. Click Save.



Step 5: Create Academic Standing Compliance Flag

1. Click Create Calculated Field.

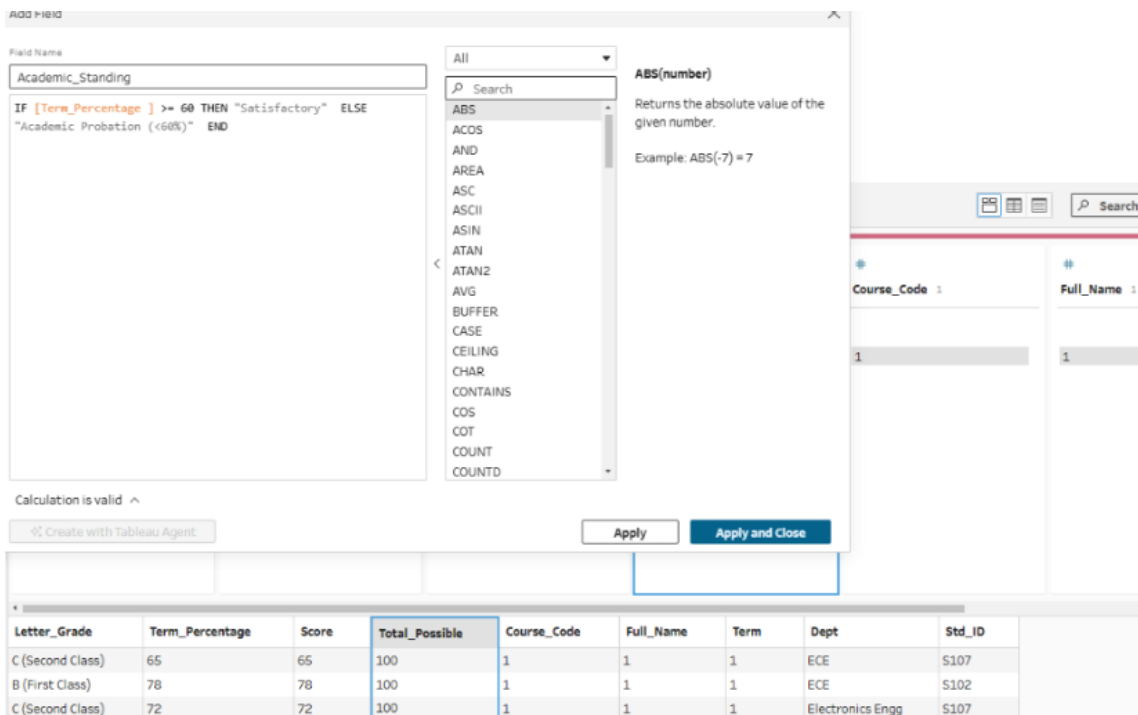
2. Field Name: Academic_Standing

3. Tableau Prep Formula:

Code snippet

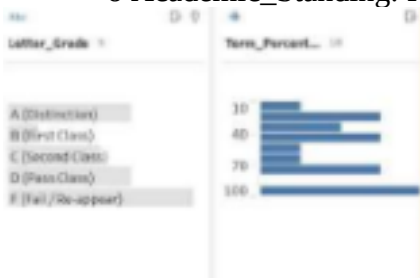
```
IF [Term_Percentage] >= 60 THEN "Satisfactory"
ELSE "Academic Probation (<60%)"
END
```

4. Click Save.



Step 6: Verify Transformed Profile Distributions

- Look at the newly generated profile cards in Clean 4:
 - Term_Percentage: Displays a smooth numeric distribution between 0% and 100%.
 - Letter_Grade: Displays discrete categorical buckets (A (Distinction), B (First Class), etc.).
 - Academic_Standing: Highlights students requiring academic counseling.



G) Create and automate data preparation flows

Step 1: Add an Output Step to the End of the Pipeline

- On the Flow canvas, locate the final cleaning step (Clean 4).
- Hover over the Clean 4 icon and click the + icon.
- Select Output (represented by the download/save arrow icon).
- An Output step node will appear connected directly to Clean 4.

Step 2: Configure Output Settings in the Bottom Pane

- Click on the Output node to open its configuration panel below:
 - Save output to: Select File (or Tableau Server / Tableau Cloud).
 - Output type: Select Tableau Data Extract (.hyper) for high-speed dashboarding, or Comma Delimited Text (.csv) for universal export.
 - Location: Click Browse and choose the directory where you want the final dataset saved.
 - File Name: Enter Student_Performance_Master_Cleaned.
 - Write Option: Select Create table (or Append to table if updating existing records).

Step 3: Run the Data Prep Flow

- In the Output pane at the bottom-left, click the blue Run Flow button (or click the play icon  directly on the Output node on the canvas).

2. Tableau Prep will process all upstream nodes (Cleaning $\rightarrow$ Grouping $\rightarrow$ Unioning $\rightarrow$ Merging $\rightarrow$ Aggregating $\rightarrow$ Scoring).
3. A success notification will appear stating:
"The flow ran successfully in X seconds."



Step 4: Save and Automate the Flow File

1. Go to the top menu bar $\rightarrow$ select File $\rightarrow$ Save As....

2. Name your workflow:

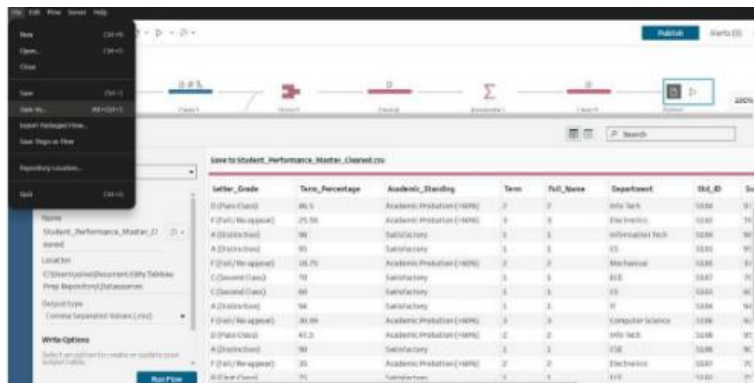
Plaintext

Student_Performance_Prep_Flow.tfl

3. (Automation Options in Production):

- o Command Line Execution: You can automate this flow on Windows/Mac using the Tableau Prep Command Line Interface (CLI): `Bash tableau-prep-cli -t Student_Performance_Prep_Flow.tfl`

- o Tableau Prep Conductor: If connected to Tableau Cloud/Server, you can publish the flow and set a scheduled refresh (e.g., every Monday at 6:00 AM) to automatically run whenever new semester files are added.



Step 5: Verify the Final Master Dataset in Tableau Desktop

1. Open Tableau Desktop.
2. Connect to the exported Student_Performance_Master_Cleaned.csv (or .hyper).
3. Build the self-performance dashboard using the pre-calculated

